

JEE MAIN REVISION MASTER: ROTATIONAL MOTION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 247 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Rotational Motion - Core Application	247 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
-----------------	--

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Rotational Motion - Core Application (Very High Yield — 247 PYQs)

Core Principle: Key problem pattern in Rotational Motion.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A solid cylinder having radius R and length L is slipping on a rough horizontal plane. At time $t=0$ the cylinder has a translational ve..." [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "An object of uniform density rolls up the curved path with the initial velocity v_o as shown in the figure. If the maximum height attain..." [Ans: D]
- [2026 Morning Shift] Q#1 (MCQ): "Two circular discs of radius each 10 cm are joined at their centres by a rod of length 30 cm and mass 600 gm as shown in figure. If the mass..." [Ans: D]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A solid cylinder having radius R and length L is slipping on a rough horizontal plane. At time $t=0$ the cylinder has a translational velocity $v_0=49\text{m/s}$, perpendicular to its axis and a rotational velocity $\omega_0=v_0/4R$ about the centre. The time taken by the cylinder to start rolling is _____ seconds. (coefficient of kinetic friction $\mu_k=0.25$ and $g=9.8\text{m/s}^2$)

- (A) 15
- (B) 5 **[Correct]**
- (C) 10
- (D) 7.5

Correct Answer: B

Solution Analysis: The cylinder is slipping, which means its translational motion and rotational motion are not in order ($v=r\omega$) for pure rolling. Initial translational velocity is $v_0=49\text{m/s}$. Initial angular velocity is $\omega_0=\frac{v_0}{4R}$, where R is the radius of solid cylinder of length L . For pure rolling to start, the condition $v=\omega R$ must be satisfied. Initially, $\omega_0 R=\frac{v_0}{4}$, which is less than v_0 , so the cylinder is sliding forward. Since the cylinder is sliding forward, kinetic friction (f_k) acts backward at the point of contact. $f_k=\mu_k$